

Proposal Feedback: AV Crash Severity and Pattern Analysis

Student Names: Santiago Oscar Aramayo Velasco Sr., Lauren McDonald, Luis Velez

Overall Alignment with ISYE 4600 Guidelines

- The proposal defines a high-stakes, IE-relevant problem with clear stakeholders (AV safety engineers and regulators) and explains why predictive analysis improves decision-making.
- The dataset is clearly identified, with source, unit of analysis, size, variables, and severity definition explicitly stated.
- The methodology includes a clear baseline (logistic regression), stronger comparison models (random forest, gradient boosting), and an unsupervised component for pattern discovery.
- The evaluation plan appropriately addresses class imbalance and defines measurable success criteria.

Suggestions for Improvement for Final Submission

1. Clarify the exact definition of the severity label (e.g., logical rule combining injury, towing, airbag deployment) and justify why this operationalization reflects meaningful safety risk.
2. Explicitly describe the train/test split strategy (e.g., temporal split vs. random split). Given policy implications, a temporal validation approach may better simulate deployment.
3. Address potential reporting bias and data limitations (e.g., underreporting, differences between Level 2 and ADS reporting requirements) and discuss how this affects generalization.
4. Clearly define the majority-class baseline in addition to logistic regression to strengthen performance comparisons.
5. In the final report, include structured error analysis focusing on false negatives (missed severe crashes) and explain the operational consequences.
6. For clustering, justify feature selection and the choice of number of clusters, and clearly explain how clusters translate into actionable crash “scenarios.”
7. Strengthen the “Interpretation for the System” section by translating model findings into concrete safety prioritization recommendations.